

PROBLEM SET 3

(Due Feb 3, 2023 @ 8:30 AM in Gradescope and Moodle)

This problem set is to be completed **individually** and **entirely in Excel**. No hardcopies are needed for Problem 1 - Problem 3. You will submit your assignments via Moodle ([screenshots to Gradescope](#)). Make sure you start a blank workbook on your own computer. Solve each problem on a separate Spreadsheet; name the Spreadsheets as “Problem 1”, “Problem 2”, “Problem 3”.

Problem 1 (45 points total)

Here is a problem based on experimental data from the Genzer research group. A polymer coating can be generated by grafting one end of polymer chains to a solid support (typically glass-like substrate). The systems are characterized by the length of the polymer chains (*i.e.*, their molecular weight) and the density of the grafts on the substrate (so-called grafting density, σ , defined as the number of chains per unit area of the substrate surface). Due to various factors (both reaction related as well as crowding of chains as they grow away from the substrate), the length of the polymer grafts may vary considerably (see left cartoon in **Figure 1**). We can measure the distribution of the molecular weights by using a technique called size exclusion chromatography (SEC). However, we need to first remove the polymers from the substrate before we pass them through the SEC apparatus (see more details in References 1,2, 3 and 4). A chemical called tetrabutyl ammonium fluoride (TBAF) is used to selectively cleave Si-O bonds, which attach the polymers to the substrate. We monitor the progress of degrafting over time and obtain values of the grafting density of the remaining polymers at each instant (in **Figure 1**, the degrafted chains are marked in light blue and the chains remaining on the surface are in dark blue).

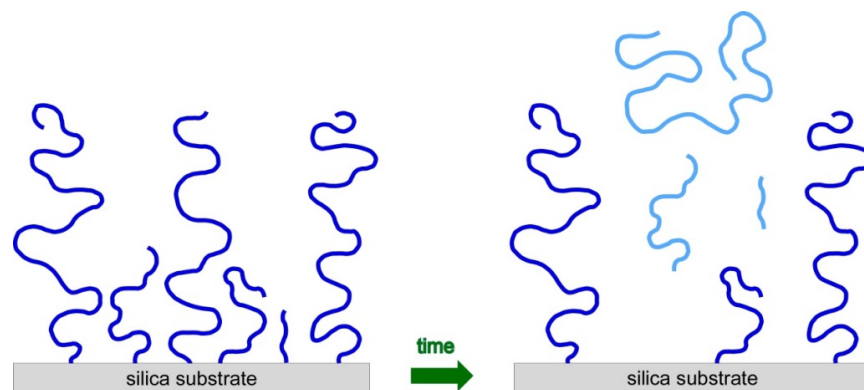


Figure 1. Schematic depicting degrafting of macromolecules from solid substrates. The grafting density of polymer grafts on the surface (σ) decreases with increasing time.

¹ R.R. Patil, S. Turgman-Cohen, J. Šrogl, D. Kiserow, J. Genzer, “*ACS Macro Letters* **5**, 251-254 (2015).

² R.R. Patil, S. Turgman-Cohen, J. Šrogl, D. Kiserow, J. Genzer, *Langmuir* **31**, 2372-2381 (2015).

³ R.R. Patil, D. Kiserow, J. Genzer, *RSC Advances* **5**, 86120-86125 (2015).

⁴ R. Patil, J. Miles, Y. Ko, P. Datta, B. Rao, D. Kiserow, J. Genzer, *Macromolecules* **51**, 10237-10245 (2018).

Table 1 provides values of the grafting density as a function of degrafting time. We will operate with these real data to answer a few questions.

- a. (10 points) Assume that the degrafting process can be described by the 2nd order reaction kinetics (you will learn more about reaction kinetics in CHE 446):

$$\sigma = \frac{\sigma_0}{1 + k \cdot t \cdot \sigma_0} \quad (1)$$

In Equation (1), k is the rate constant, which describes the kinetics of the reaction, σ_0 is a constant (whose physical meaning you will be asked to answer in part d), t is the degrafting time. [Note: in reality, the reaction is likely 1st order, but we can ignore this fact for the sake of this exercise; we thus stick with the 2nd order reaction kinetics].

Degrafting time (min)	σ (chains/nm ²)
0.5	0.493
6	0.401
10	0.399
14	0.358
20	0.283
30	0.234
40	0.202
50	0.169
60	0.145
70	0.131

Linearize the expression given in Equation (1) and plot the data in an Excel sheet. Obtain the values of the slope and the intercept using the Trendline, and calculate the corresponding values of the constants σ_0 and k .

- b. (10 points) Repeat the same procedure as in part a) except now use the function LINEST to obtain the slope and the intercept.
- c. (10 points) Using Solver fit the data in **Table 1** directly to the function given by Equation (1) to obtain the values of σ_0 and k .
- d. (5 points) What is the physical meaning of σ_0 ?
- e. (5 points) Can you guess what is the physical meaning of k ? [Hint: look at the unit of k and you may guess the answer]. Assume that you increase the rate of the degrafting reaction (either by increasing temperature or the concentration of the degrafting TBAF agent). Would you expect k to increase or decrease with increasing the rate of the degrafting reaction?
- f. (5 points) If the values you calculated in part c) are different from those you obtained in parts a) and b), provide at least *two* brief explanations for the discrepancy. Which of the two approaches (a/b or c) do you think is a better fit of the data and why?

Problem 2 (25 points total)

[from Wiki] Moore's law describes a long-term trend in the history of computing hardware, in which the number of transistors that can be placed inexpensively on an integrated circuit has doubled approximately every two years. Check the entry on Moore's law on Wiki to learn more about this phenomenon.⁵

⁵ https://en.wikipedia.org/wiki/Moore's_law

- a. (15 points) Use the data set in **Table 2** to predict the number of transistors in year 2012 using i) Trendline and ii) functions SLOPE and INTERCEPT in Excel. Assume that the data follow an **exponential trend**, *i.e.*, $T = a \cdot \exp(b \cdot Y)$, where a and b are constants and T and Y are the numbers of transistors and year, respectively. For approach ii) you would have to linearize the data. You can linearize the data for approach i) as well, if you wish.
- b. (10 points) Prove that the number of transistors approximately doubles every two years. In answering this part, do not use the individual data from **Table 2**. Instead, use the function you obtained in part a). (Hint: use ratio of $T(Y+2)/T(Y)$)

Table 2		
Processor	Year (Y)	Transistors (T)
8008	1972	2.5
8080	1974	5
8086	1978	29
80286	1982	120
80386	1985	275
80486	1989	1,180
Pentium	1993	3,100
Pentium II	1997	7,500
Pentium III	1999	24,000
Pentium 4	2000	42,000
Itanium 2	2003	220,000
Itanium 2	2004	592,000

Problem 3 (30 points total)

Consider population growth in 3 countries show in **Figure 3** (data taken from Reference 6).

- a. (10 points) Use the data in **Figure 3** (also in **Table 3**) to comment on the trends in the population in each of the countries between 1500-1800s, 1800-1900s, and 1900-2000s. What are the potential socio-economic reasons for the population variation for each of the countries? (Hints: consider the land size of each country, economical standing, migration in and out, the time of foundation of each country (if applicable), *etc.*)

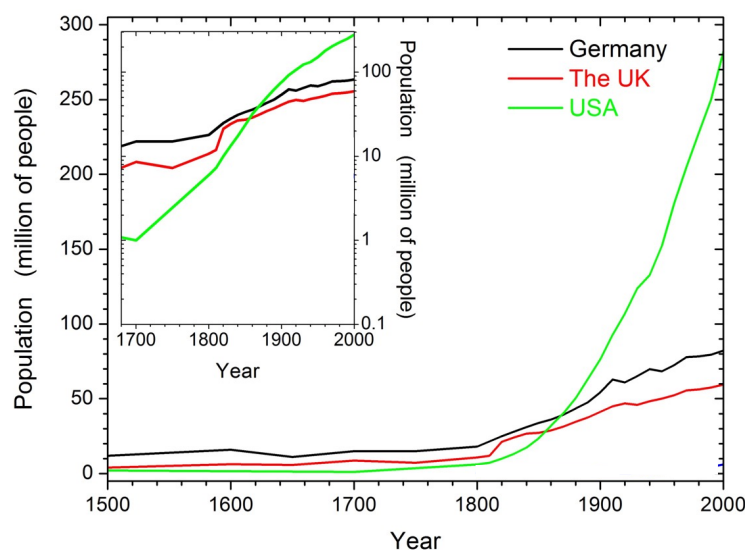


Figure 3. Population (ordinate, *i.e.*, y-axis) as a function of year (abscissa, *i.e.*, x-axis) for Germany, the UK and the US. The inset depicts a semi-logarithmic plot of the same data set – concentrating on the past 3 centuries.

- b. (20 points) Taking the data in **Table 3**, estimate when would the US reach the 300 M population mark (Well, it has already reached more than a decade ago)⁶. Further, use the data to predict the number of people living in the US in 2019. Try at least **three different functions** to fit the data, such as **linear**, **exponential**, and **power** function. Use the linearization method to find the constants for each method (skip the first three data points (Year 1500, 1600, and 1700) if needed). How does this estimate compare to the value reported officially? (328.3 million in 2019, based on the Internet). Is there a discrepancy? If so, comment on a potential source (or potential sources) thereof.

Table 3

Year	Population		
	Germany	The UK	USA
1500	12000000	3942000	2000000
1600	16000000	6170000	1500000
1650	11000000	5800000	
1700	15000000	8565000	1000000
1750	15000000	7250000	
1800	18000000	10750000	6000000
1810		11970000	7240000
1820	24905000	21239000	9980510
1830	28045000	24139000	13240314
1840	31126000	26745000	17443768
1850	33746000	27181000	23579718
1860	36049000	28888000	31838901
1870	39231000	31400000	40240630
1880	43500000	34623000	50458000
1890	47607000	37485000	63302000
1900	54388000	41155000	76391000
1910	62884000	44916000	92767000
1920	60894000	46821000	106881000
1930	65084000	45866000	123668000
1940	69835000	48226000	132637000
1950	68374572	50127000	152271000
1960	72480869	52372000	180671000
1970	77783164	55632000	205052000
1980	78297904	56314000	227726463
1990	79380394	57493307	250131894
2000	82187909	59522468	282158000

⁶ <http://www.today.com/id/15298443/ns/today/t/us-population-hits-million-mark/>